

Program : Diploma in Tool & Die Engineering	
Course Code : 3116	Course Title: Tool Engineering Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To introduce basic engineering processes for manufacturing and assembly.
- To familiarize students with the identification, selection & use of various marking, measuring, holding, striking, and cutting tools and equipment's.
- To provide knowledge to interpret job drawings, produce jobs, and inspect the job for specified dimensions.
- To impart skill to operate and control different machines by adopting safety practices.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Bench work operations		Production Engineering I	2
Identification of lathe parts		Production Engineering II	3
Basic filing operations		Engineering Workshop practice	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Produce skills in basic bench working practice to identify, select and use various marking, measuring, holding, striking and cutting tools, equipment's & machines.	15	Applying
CO2	Demonstrate drilling and reaming operation.	6	Applying

CO 3	Demonstrate the operation of tap & Die set.	6	Applying
CO4	Practice various operations on tool room lathe and pedestal grinding machine.	15	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						3
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Produce skills in basic bench working practice to identify, select and use various marking, measuring, holding, striking and cutting tools, equipment & Machines.		
M1.01	Practice Filing of flat surfaces and right angles with try square accuracy.	5	Applying
M1.02	Practice filing of flat and parallel surfaces within +/- 0.5mm accuracy along with marking, punching, hack sawing and chamfering at 450	5	Applying
M1.03	Practice filing & finishing of angles within +/- 10 and checking angles using bevel gauges.	5	Applying
M1.04	Practice filing of external radius and marking with radius gauge.		Applying
CO2	Demonstrate drilling and reaming operation.		
M 2.01	Illustrate drilling machine, drilling through holes, chain drilling and parting off, drilling blind holes.	3	Applying
M 2.02	Demonstrate radial drilling machine operations.		Applying

M 2.03	Perform counter boring and counter sinking operation.	2	Applying
M 2.04	Practice reaming with hand reamers	1	Applying
	Lab Exam – I	1.5	
CO3	Demonstrate the operation of tap & Die set		
M3.01	Explain the tap hole size calculation	3	Understanding
M3.02	Perform tap and die operation on different Metric threads.	3	Applying
CO4	Practice on tool room lathe and pedestal grinding machine		
M4.01	Identify various operations on tool room lathe and pedestal grinding machine.	1	Understanding
M4.02	Practice grinding of single point cutting tools to required angle using pedestal grinding machine.	1	Applying
M4.03	Practice various turning Exercises (i) Facing, Step Turning & Chamfering (ii) Step Turning & Taper Turning (iii) Step Turning & Groove Cutting (iv) Step Turning & Knurling (v) Step Turning & Thread Cutting (vi) Turning and Drilling.	13	Applying
	Lab Exam - II	1.5	

Text/ Reference:

T/R	Book Title/Author
T1	Mechanical Workshop Practice by K. C Jon, PHI Learning Private Limited
R1	Mechanical Workshop & Laboratory Manual by K. C. John
R2	Workshop Technology Vol. I by S K HajraChoudhary
R3	Workshop Technology Vol. II by S K HajraChoudhary

Online Resources:

Sl.No	Website Link
1	http://ecoursesonline.iasri.res.in/mod/page/view.php?id=98833
2	https://www.profitfromtraining.org.uk/material__fabrication__tools_and_measuring_workbook.pdf
3	http://www.survivorlibrary.com/library/machine_tools_and_workshop_practice_1905.pdf
4	https://www.engineeringbookspdf.com/mechanical-engineering/workshop-practice/
5	https://www.agrimoon.com/wp-content/uploads/WORKSHOP-PRACTICE.pdf